

Programming with Python MID

Show equations, substitutions, intermediate reasoning, and a final labeled answer.

1. [6 pts] Trace `out = [i*i for i in range(4) if i % 2 == 0]`. Give the resulting list and explain which values are filtered.

2. [8 pts] Write a function `first_unique(xs)` that returns the first value appearing once, or None if every value repeats. State its time complexity.

3. [7 pts] The loop `for i in range(len(xs)-1): if xs[i] > xs[i+1]: swap` is intended to bubble the maximum to the end. Identify the missing condition or pass logic and justify the fix.

4. [6 pts] Compare a binary-search lookup and a linear scan on $n=64$ sorted records. Give worst-case comparisons and asymptotic bounds.

5. [5 pts] Explain why mutating a list while iterating over it can skip elements. Give a safe alternative.

6. [6 pts] Trace `out = [i*i for i in range(4) if i % 2 == 0]`. Give the resulting list and explain which values are filtered.

7. [8 pts] Write a function `first_unique(xs)` that returns the first value appearing once, or None if every value repeats. State its time complexity.

8. [7 pts] The loop `for i in range(len(xs)-1): if xs[i] > xs[i+1]: swap` is intended to bubble the maximum to the end. Identify the missing condition or pass logic and justify the fix.
